



Exercise-induced skeletal muscle signaling pathways and human athletic performance



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ABSTRACT

Skeletal muscle is a highly malleable tissue capable of altering its phenotype in response to external stimuli including exercise. This response is determined by the mode, (endurance- versus resistance-based), volume, intensity and frequency of exercise performed with the magnitude of this response-adaptation the basis for enhanced physical work capacity. However, training-induced adaptations in skeletal muscle are variable and unpredictable between individuals. With the recent application of molecular techniques to exercise biology, there has been a greater understanding of the multiplicity and complexity of cellular networks involved in exercise responses. This review summarizes the molecular and cellular events mediating adaptation processes in skeletal muscle in response to exercise. We discuss established and novel cell signaling proteins mediating key physiological responses associated with enhanced exercise performance and the capacity for reactive oxygen and nitrogen species to modulate training adaptation responses. We also examine the molecular bases underpinning heterogeneous responses to resistance and endurance exercise and the dissociation between molecular ‘markers’ of training adaptation and subsequent exercise performance.

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1. Introduction

The conversion of multiple signals generated during exercise to molecular events aimed at conserving cellular homeostasis and ultimately inducing phenotypic changes in skeletal muscle involves a cascade of events resulting in the activation and/or repression of specific signaling pathways regulating gene expression and protein synthesis/degradation [1–4]. One exercise-induced perturbation to cellular homeostasis is the increase in reactive oxygen (ROS) and nitrogen (RNS) species [5]. The generation of ROS and RNS products by the mitochondria and other subcellular compartments with exercise induce cellular damage and activates redox signaling pathways that can modulate the molecular mechanisms regulating protein synthesis and breakdown processes that ultimately form the basis for exercise training adaptations [5,6].

An important concept developed over the past decade is that the chronic responses to exercise training are likely to be the result of the acute, but cumulative effects of the responses to single exercise bouts [7]. As such, these acute and transient changes in gene

transcription following a single exercise bout, when reinforced by repeated exercise stimuli, result in chronic effects on the rates of protein breakdown/synthesis that ultimately form the basis of skeletal muscle training adaptation and improvements in exercise capacity/performance [2,8]. Yet despite major breakthroughs in our understanding of how different exercise modalities activate specific cellular, molecular, and biochemical pathways, our understanding of how these effects exert their performance-enhancing benefit remains elusive. This is, perhaps, not surprising, given that exercise performance on any given day is ultimately the result of integrating multiple physiological, biomechanical and psychological factors simultaneously under a variety of different environmental conditions. Indeed, some of the variability observed in the physiological responses to standardized training protocols is likely to be underpinned by the multi-factorial and complex nature of the ‘exercise response.’ In this review we examine the molecular basis for exercise training-induced adaptations in skeletal muscle in response to both resistance- and endurance-based exercise including the roles of ROS and RNS on these cellular processes. We also examine the molecular bases underpinning these adaptations that may help explain the heterogeneous responses to exercise training, and the apparent dissociation between molecular ‘markers’ of training adaptation and subsequent exercise performance. The reader is also referred to several recent reviews published on these topics [2,3,9].

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2. Molecular mechanisms mediating adaptation to resistance exercise

Mechanical overload of skeletal muscle promotes an increase in myofiber cross-sectional area, a process termed hypertrophy [10]. Resistance-based exercise (REX) provides the optimal “anabolic” signal to stimulate the protein synthetic response, with increases in muscle protein synthesis (MPS) and net myocellular protein accretion underlying the growth in individual muscle fibers and total muscle cross-sectional area [11]. REX elevates rates of MPS above basal levels for at least 24 h [12–15], but there is also a small rise in muscle protein breakdown [12]. Thus, any training-induced hypertrophic response requires a net accretion of contractile myofibrillar protein.

REX-induced increases in MPS are largely attributed to upregulation of protein translation initiation and control by the mechanistic target of rapamycin (mTOR) serine/threonine protein kinase [16]. mTOR exists in two multi-protein complexes (mTORC1 and mTORC2) with mTORC1 the predominant regulator of translation initiation. Two downstream substrates, ribosomal protein s6 (rps6) p70 kinase (p70^{S6K}) and eukaryotic initiation factor 4E (eIF4E)-binding protein 1 (4E-BP1) orchestrate a number of signals

from mTORC1 (Fig. 1). p70^{S6K} has several substrates that contribute to various steps of protein translation and has recently been implicated in the transcriptional regulation of ribosome biogenesis (i.e., to augment protein translational capacity) [17], whereas 4E-BP1 phosphorylation is thought to predominantly lead to the translation of 5'-tract of pyrimidine (5'-TOP) mRNAs that encode for translation factors and ribosomal proteins [18]. While some associations between p70^{S6K} phosphorylation and acute post-REX MPS [19,20] and training-induced hypertrophy [21–24] exist, larger p70^{S6K} phosphorylation responses have also been shown to parallel REX volume and/or lifting intensities [25–28] with the most pronounced effects in type II (fast-twitch) muscle fibers [29,30]. This latter point should not be dismissed, as a single bout of maximal eccentric contractions elicited a larger p70^{S6K} phosphorylation than that provoked by maximal concentric efforts [31]. These findings may (a) reveal a potential molecular basis for the efficacy of eccentric-based REX training in promoting type II fiber hypertrophy [32] or (b) the higher force imposed by eccentric versus concentric work is merely a larger homeostatic perturbation within susceptible fibers (i.e., type II that rely heavily on fast glycolytic energy yield) and ultimately is inconsequential for chronic training adaptations [33,34]. In support for the latter

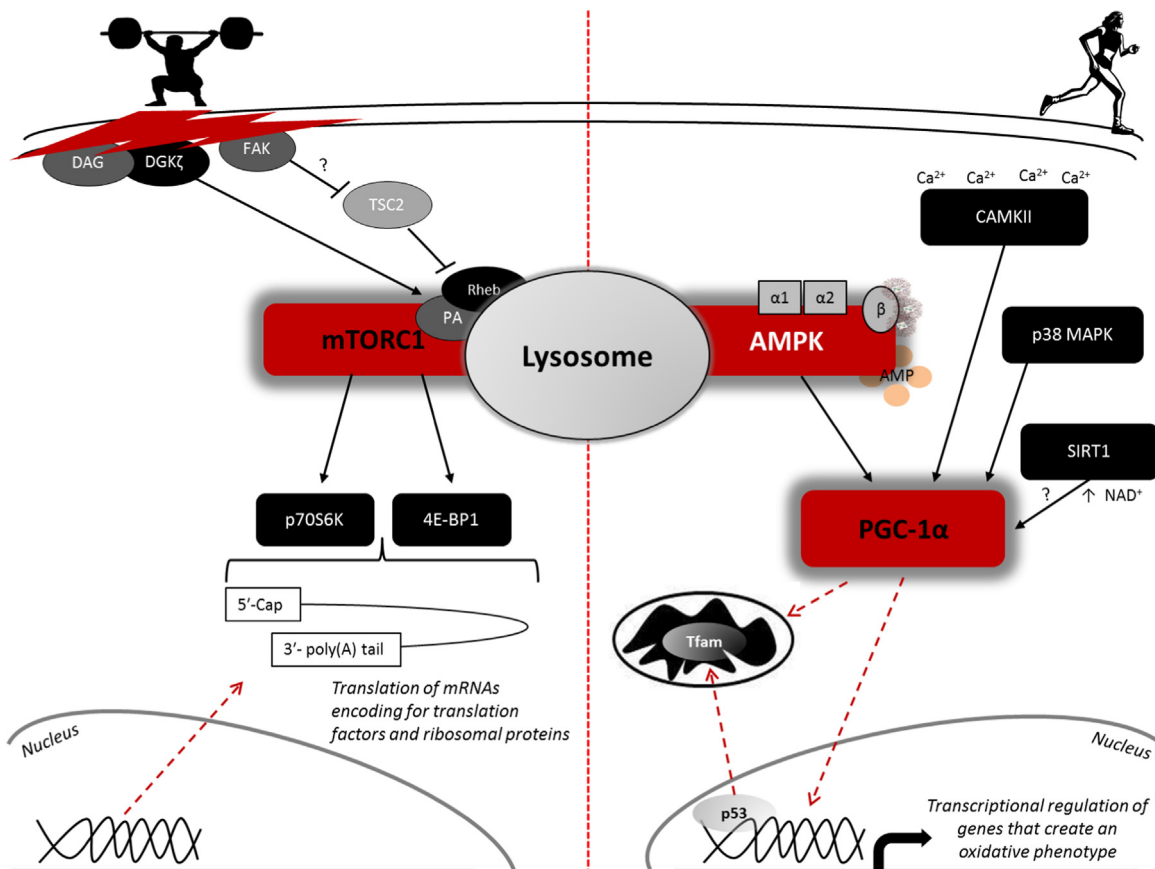


Fig. 1. Several signals propagated by resistance- and endurance-based exercise contraction converge on the lysosome. High-force resistance exercise (REX) contractions perturb transmembrane structures leading to the accumulation of membrane diacylglycerol (DAG), increased DAG kinase ζ (DGK ζ) isoform activity and phosphatidic acid (PA) synthesis that, in turn, binds directly to mTORC1 in the same region as the inhibitory rapamycin-binding domain. In addition, REX triggers tuberlin (or TSC2) removal from the lysosome enabling mTORC1 to interact with Ras homolog enriched in brain (Rheb), a small GTPase that is negatively regulated by TSC2. Sarcomeric adhesion-associated signaling molecules, such as focal adhesion kinase (FAK), may stimulate mTORC1 activity by similar mechanisms of TSC2-lysosomal abrogation. The collective outcome is an increase in cap-dependent mRNA translation and increases in cell size that is mitigated primarily by mTOR kinase substrates, p70 ribosomal protein s6 kinase (p70^{S6K}) and eIF4E binding protein 1 (4E-BP1). The endurance exercise (END)-evoked disruptions to cell homeostasis activates several protein kinases that seemingly “sense” the degree of stress imposed. For example, END triggers calcium (Ca^{2+}) oscillations to sustain locomotion (Ca^{2+} -dependent protein kinase II; CAMKII), causes glycogen and ATP depletion (AMP-dependent kinase; AMPK), elevates oxidative stress (p38 mitogen-activated kinase; p38 MAPK) and alters the reduction/oxidation state (sirtuin 1; SIRT1). These signals converge to post-translationally modify PPAR γ -coactivator 1 α (PGC-1 α) which, in the compensatory defense of homeostasis, promotes an oxidative phenotype by coactivating numerous nuclear- and mitochondrial-encoded genes. Black arrows denote activation; straight black lines denote inhibition and dashed red arrows denotes translocation. Question marks refers to findings that are either equivocal or have not been confirmed *in vivo* skeletal muscle.

contention, $p70^{S6K}$ activity declines with training [35–37] and becomes activated/deactivated more rapidly after REX in the trained muscle [15]; indicative of a greater efficiency of restoring homeostasis [36].

The precise upstream kinases that stimulate the initial rise in mTOR activity remain largely unknown, although the lysosome, a digestive organelle that integrates metabolic cues to maintain cellular energy balance [38], has been identified as a key site stimulating mTOR translational responses to resistance-like mechanical stimuli [39]. At the lysosome, mTOR is directly activated by Rheb, a small GTPase whose affinity for mTOR is controlled by its nucleotide-binding state and negatively regulated by tuberlin (or TSC2) [40,41]. Recently, Jacobs et al. [42] demonstrated that eccentric contractions in mice induced phosphorylation of TSC2, triggering its lysosomal dissociation to promote mTOR–Rheb interactions. It is likely that the mechanically-induced lysosomal removal of TSC2 is independent of canonical growth factor/Akt signaling [43–47] and orchestrated by a *hitherto* unknown mechanoreceptor(s). This latter notion may explain why creating an elevated systemic hormonal milieu of insulin-like growth factor-1 (IGF-1), among other reputedly “anabolic” hormones, fails to confer a synergistic effect toward REX-induced mTORC1 signaling, MPS [20] and training-induced hypertrophy in humans [48]. In addition, the AMP-activated protein kinase AMPK, which also localizes to the lysosome for activation following energy stress [49], stimulates TSC2 activity under similar stress conditions [50]. Although the inhibitory function of AMPK toward MPS following acute REX is equivocal [51], these data at least suggest that in response to mechanoreceptor stimulation, the prevailing anabolic response (involving a complete repression of TSC2) is only optimized when sufficient intracellular energy is available [52].

A second direct activator of mTOR is phosphatidic acid (PA), a glycerophospholipid that competes with rapamycin for access to mTOR [53]. Mechanically-induced activation of mTOR increases intramyocellular concentrations of PA in rodent models [46,54,55] and this is thought to primarily be the result of increased PA synthesis from diacylglycerol (DAG) by DAG kinase ζ isoforms (DGK ζ) [55]. Of note is that increasing the duration of mechanical stretch increased the amount of membrane-associated DGK ζ [55]. The ability for sustained mechanical tension to facilitate this DGK ζ membrane-activity and subsequent PA synthesis is unknown, but recent evidence reveals that cell damage elevates DAG concentrations at the damaged cytoskeletal site to facilitate its repair and promote membrane fusion [56,57]. Considering the focal site of skeletal muscle force transmission are the adhesive structures (i.e., transmembrane integrins) that connect the sarcomeric Z-disc to the extracellular matrix [58], it is tempting to speculate that mechanically-invoked perturbations to this region triggers DAG accumulation, thereby providing substrate for PA synthesis and mTORC1 signaling. While this notion may explain how eccentrically-induced muscle-damage might provoke PA-mTORC1 activity, not all REX bouts elicit muscle damage and alternative mechanisms are likely to transduce mechanical tension to mTORC1 activation, such as through adhesion site-associated focal adhesion kinase/TSC2 signaling [59].

Few studies have investigated the effects of ROS and RNS on mTOR-related signaling and muscle hypertrophy responses following resistance-based exercise. This is most likely due to exercise-induced ROS being a by-product of increased oxidative metabolism whereas the low duration, high-intensity intermittent nature of resistance exercise are fueled predominantly by oxygen-independent fuel sources. Nonetheless, recent studies in rodent [60] and human [61,62] skeletal muscle demonstrate attenuated Akt, mTOR, ribosomal protein S6, p38 MAPK, ERK1/2 and $p70^{S6K}$ phosphorylation with anti-oxidant (vitamin C, E and N-acetylcysteine) supplementation following both a single bout and

chronic resistance exercise training. While the molecular mechanisms responsible for this reduction in ‘anabolic’ cell markers with antioxidants remain unknown, it is purported that reduced neuronal nitric oxide synthase activation of transient receptor potential cation channel, subfamily V, member 1 (TRPV1) may play a role [63]. However, Paulsen and colleagues reported similar increases in muscle mass and rates of muscle protein synthesis between individuals supplementing with either Vitamin C and E or a placebo following 10 weeks of resistance training despite acute decreases in $p70^{S6K}$ and p38 MAPK phosphorylation [62]. This discrepancy in cell signaling and muscle hypertrophy responses may be a result of an initial cellular response induced by a novel stimulus (i.e., antioxidants) that is slowly reversed overtime as the cell desensitizes to its presence. Further studies investigating different antioxidant dosage strategies (i.e. amount, type, length of consumption, etc.) are required to better understand the role of ROS and other nitric oxide derivatives on strength and hypertrophy responses with resistance exercise.

3. Molecular mechanisms mediating adaptation to endurance exercise

Endurance exercise (END) enhances whole-body maximum oxygen uptake ($\dot{V}O_{2max}$) and muscle oxygen extraction (via increased blood flow). These adaptations largely underpin the enhanced ability of skeletal muscle to oxidize both carbohydrate- and fat-based fuels after chronic endurance training [64]. Mitochondrial biogenesis underpins skeletal muscle oxidative capacity and is therefore considered the hallmark intracellular adaptation to END. Specifically, END perturbs the intracellular milieu by accelerating bioenergetics and redox reactions [i.e., raising AMP/ADP, inorganic phosphate (P_i) and NAD^+ concentrations], triggering respiratory bursts of free radicals and inducing calcium (Ca^{2+}) oscillations for sustained actin–myosin cross bridge cycling [1,3,65]. Hence, when repeated over days, weeks and months, muscle cells respond to these challenges by increasing mitochondrial number. Peroxisome proliferator-activated receptor- γ (PPAR γ)-1 α (PGC-1 α) is considered the “master regulator” of mitochondrial biogenesis [66,67]. PGC-1 α co-activates transcription factors that upregulate nuclear and mitochondrial genes encoding for proteins involved in respiratory chain function, substrate oxidation and angiogenesis, the latter being important for nutrient and O_2 delivery to the working muscles [68] (Fig. 1). Indeed, in rodent models where PGC-1 α has been overexpressed in skeletal muscle, there are improvements in whole-body VO_{2peak} and END performance [69] with tighter metabolic control (i.e., a shift from CHO- to fat-based fuels at the same relative exercise intensity).

Elevated PGC-1 α mRNA levels are observed following endurance-based exercise in humans [7,70–73], including highly-trained athletes [74], with the highest abundance of mRNA generally observed in the first 2–4 h post-exercise. Moreover, PGC-1 α translocates to the nucleus in human skeletal muscle following both moderate- [75] and high-intensity END [76], while its nuclear abundance [77] and total protein levels [78] can increase after just 2 weeks of interval-based END training. Of note, PGC-1 α can also localize to mitochondria following END where it forms a transcriptional complex with mitochondrial transcription factor A (Tfam) at the mtDNA D-loop [79], a region associated with respiratory capacity, mtDNA replication and transcription [80]. Another protein, p53, similarly translocates to the mitochondria following acute END in rodents to form a Tfam-bound transcriptional complex at the mtDNA D-loop [81]. Since PGC-1 α coactivates nuclear p53 to enhance mitochondrial respiration in response to metabolic stress [82], it is possible that its serves a similar function toward mitochondrial p53 following END; however this needs to

experimentally confirmed. Nonetheless, these latter points are important, as mtDNA encodes for 13 proteins of the respiratory chain, the transcripts of which (e.g., cytochrome *c* oxidase subunit I) along with mtDNA concentrations have been found to be higher in endurance-trained athletes compared to their untrained counterparts [83].

A growing body of evidence suggests ROS and RNS can modulate PGC-1 α mediated increases in mitochondrial biogenesis by altering the phosphorylation status of several key upstream activators of PGC-1 α including p38 MAPK and AMPK [6,84,85]. ROS generated by endurance-based exercise are typically maintained within physiological limits through the activity of several antioxidant enzymes [5]. These ‘physiological’ levels of ROS are thought to be part of the ‘normal’ cellular milieu regulating increased skeletal muscle mitochondrial biogenesis with endurance exercise [86]. However, increases in ROS beyond normal physiological ranges may impair muscle function and cause oxidative damage, forming the basis for potential antioxidant supplementation when exercising [6]. Early studies in rodents demonstrated antioxidant supplementation during treadmill running reduced skeletal muscle p38 MAPK and ERK phosphorylation [85], as well as PGC-1 α mRNA expression, mitochondrial protein and activity levels [87,88]. Ristow and colleagues confirmed these findings in human skeletal muscle showing decreased PGC-1 α abundance with antioxidant supplementation following 4 weeks of aerobic-based exercise [89]. Moreover, exercise-induced increases in markers of endogenous ROS defence, including superoxide dismutase 1 and 2 and glutathione peroxidase, were also reduced with the antioxidant supplementation [89], suggesting antioxidant supplementation reduces mitochondrial biogenesis responses commonly observed with endurance training. In further support of this notion, vitamin C and E supplementation prevented increases in PGC-1 α protein and cytochrome oxidase 4 expression following 10 weeks of aerobic exercise involving a combination of high-intensity interval and steady state continuous sessions [90]. However, it should be noted that not all studies observe decreases in cellular measures of mitochondrial biogenesis with antioxidant supplementation following endurance exercise [91]. While this discrepancy may be a result of different supplementation regimens (dosage and duration of supplementation) between studies, similar improvements in performance outcome measures such as $\dot{V}O_{2peak}$, 20 m shuttle test and 10 km time trial have been reported between participants supplemented with antioxidants or a placebo [90,92]. Thus, while antioxidant supplementation appears to modulate some of the acute cell signaling events mediating mitochondrial biogenesis responses, this does not appear to translate to decreased performance with chronic endurance training.

The type and intensity of endurance-based exercise can also influence PGC-1 α expression. Egan et al. [93] reported larger increases in PGC-1 α mRNA in skeletal muscle of untrained humans with energy-matched work bouts of short-duration, high- ($\sim 80\% \dot{V}O_{2peak}$) or prolonged duration, low-intensity ($\sim 40\% \dot{V}O_{2peak}$) cycling. Such an increase was associated with a greater upstream phosphorylation of AMP-dependent protein kinase (AMPK), Ca^{2+} /calmodulin-dependent protein kinase II (CamKII) and p38 mitogen-activated protein kinase (p38 MAPK), which not only are *bona fide* energy-exercise-sensitive enzymes, but also regulate PGC-1 α transcription and/or activity [94–97]. The observation that high-intensity END elicits greater increases in PGC-1 α mRNA and hence might promote mitochondrial adaptation to a greater magnitude has implications for exercise prescription at the population level. However, it should be noted that both high- and low-intensity END lead to similar improvements in whole-body $\dot{V}O_{2peak}$ [98] and exercise performance [99].

While typical post-exercise increases in PGC-1 α mRNA are attenuated with training, mtDNA, mitochondrial enzyme activity

and proteins implicated in mitochondrial remodeling (i.e., membrane fission and fusion proteins) increase [7,100]. Thus, in the untrained state, high-intensity END invokes greater perturbations in cellular homeostasis that are attenuated with continued training, coinciding with an ultimate accrual of nascent mitochondrial protein and improved respiratory capacity (i.e., tighter matching of ATP synthesis with ATP hydrolysis). This notion may reconcile the fact that no differences are evident for PGC-1 α expression and upstream kinase (i.e., AMPK and p38 MAPK) activity in well-trained runners following work-matched END of divergent intensity [71]. Furthermore, because several mitochondrial proteins have long (~ 7 days) half-lives and ~ 72 h is required before any detectable exercise-induced improvements in mitochondrial function [101], the rapidity of PGC-1 α induction may serve as a form of cellular “equilibration” to prepare this challenged environment for mitochondrial protein synthesis. Indeed, several lines of evidence highlight the ability for PGC-1 α to upregulate genes mitigating glycogen storage/conservation and buffering of END-induced lactate production [67,102–104], with these effects seemingly manifesting prior to mitochondrial biogenesis [102].

4. Novel biological processes modulating endurance exercise adaptation responses

Autophagy is the constitutive turnover of cellular organelles, protein macromolecules and polyubiquitinated aggregates. Autophagy involves the formation of double-membrane vacuoles termed autophagosomes that envelop, sequester and ultimately deliver cellular constituents to the lysosome for degradation [105]. Autophagosome biogenesis is regulated by subgroups of autophagy-related gene (Atg) proteins that coordinate several facets of autophagic flux; hence sourcing of the autophagosome double-membrane, its nucleation, expansion, closure and ultimate fusion of the vesicle with the lysosome [106] (Fig. 2). Autophagy is required for basal skeletal muscle homeostasis [107] with various human muscle disorders characterized by defective autophagy responses [108]. Energy restriction is a putative autophagic stimulus, whereby limiting exogenous energy availability stimulates an increased, compensatory autophagic flux to degrade bulk portions of cytoplasm (termed macroautophagy) and release the degraded components for the maintenance of protein synthesis and thus, homeostasis [105].

Given that exercise represents a major challenge to cellular homeostasis, it is not surprising that autophagy has been implicated as a potentially important response-adaptation process [109]. Indeed, early studies in rodents showed that exhaustive running increased the number of post-exercise autophagic vacuoles surrounding damaged mitochondria [110], a selective form of autophagy termed mitophagy. Subsequent investigations in both rodents [111–117] and humans [118–121] have highlighted the sensitivity of autophagy to exercise and suggest that END, as opposed to REX [122–124], is the most potent signal driving contraction-stimulated autophagy. Indeed, autophagy seems to be involved in the protection of skeletal muscle against mechanically-induced (i.e., that which is provoked by high-force REX contractions) muscle damage [125]. Nonetheless, a recent investigation demonstrated ROS production and the prevailing hypoxic cellular environment invoked by endurance-based exercise in mice was an effective stimulus to activate autophagy [126]. ROS formation was regulated by regulated in development and DNA damage-response 1 (REDD1) forming a complex with the pro-oxidant protein TXNIP that in turn prevented hyperactivation of the cysteine protease Atg4b, thereby repressing Atg4b-mediated delipidation of LC3b-II and inhibition of autophagosome biogenesis [126]. Taking advantage of *Redd1* knockout mice, the authors revealed that

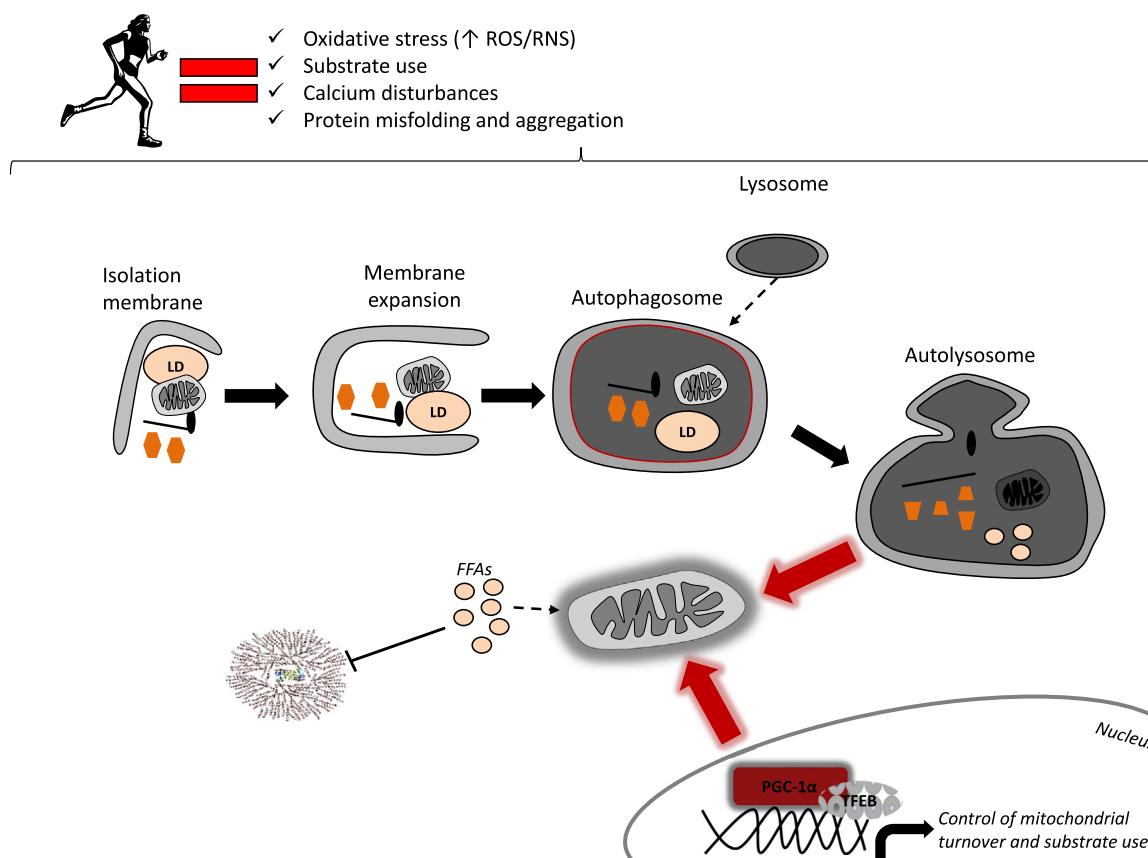


Fig. 2. Endurance exercise causes several metabolic perturbations to cellular homeostasis (noted in the schematic), all of which may stimulate an increased, compensatory autophagic flux. Autophagy commences with the selective or non-selective envelopment of a portion of cytoplasm, components of which may include (but are not limited to) lipid droplets (lipophagy), mitochondria (mitophagy) or bulk protein aggregates (macroautophagy). These cellular components are ultimately sequestered by double-membrane vesicles termed autophagosomes in a process carefully orchestrated by several autophagy-related gene (*Atg*) proteins. Upon autophagosomes fusing with the lysosomes (forming the autolysosome), the autophagic “cargo” is degraded by lysosomal hydrolases and returned to the cytoplasm as their reduced constituents. For example, autophagic degradation of lipid droplets and subsequent partitioning of free fatty acids (FFAs) to the mitochondria may support lipolysis during prolonged END, thus relieving glycogenolysis and consequent lactate accumulation. In addition, removal of long-lived and/or defective mitochondria by mitophagy (e.g., as provoked by their contraction-induced ROS emissions) precedes mitochondrial biogenesis, whereby breakdown of these mitochondria and presumably other intracellular substrates would provide the necessary energy for ultimate mitochondrial protein synthesis. These undulations in breakdown and synthesis are appropriately controlled by an autoregulatory loop of PGC-1 α /TFEB signaling, whereby each molecule can transcriptionally activate the other.

defective autophagic clearance of mitochondria attenuated END performance as a consequence of dysregulated mitochondrial function, presumably due to attenuated turnover of long-lived organelles and development of impaired oxidative phosphorylation and ATP synthesis [126]. These findings highlight the important role that physiological stresses (i.e., ROS) provoked by strenuous exercise have in stimulating the activity of constitutive pathways, such as mitophagy, that mitigate exercise-imposed disruptions to homeostasis and ultimately adaptation to the stress. Because mitophagy has been shown to precede mitochondrial biogenesis [127], it is conceivable that prior to an END-training-induced increase in nascent mitochondria (during which time the cellular environment would be otherwise poorly-equipped to deal with the stress imposed), their turnover by mitophagy must first commence. Furthermore, considering autophagy “recycles” degraded substrates, the energy released from ROS-induced mitophagy could be harnessed to supply substrate for subsequent, energy-consuming protein synthesis. Indeed, in another recent study, utilizing mice deficient for the *Atg6* gene that encodes for the protein Beclin-1, a proximal regulator of autophagosome biogenesis, Lira et al. [115] observed lower muscle capillarization, mitochondrial protein levels and no improvement in endurance performance following training. Of note was that muscle-specific overexpression of PGC-1 α enhanced basal autophagic flux and mitophagy [115]. Further support for a role for PGC-1 α in the

regulation of exercise-induced autophagy comes from whole-body PGC-1 α knockout mice who present with impaired post-exercise *Atg* transcript synthesis, mitophagy and the abundance of NPC1, a protein involved in lysosomal lipid trafficking [114]. Transcription factor EB (TFEB), the master regulator of lysosome biogenesis that translocates to the myonucleus following running exercise in mice [128], has been shown to transactivate PGC-1 α and *PPAR* α genes to exert global control over lipid catabolism in response to a starvation stimulus [129]. TFEB also upregulates genes involved in fatty acid transport and mitochondrial β -oxidation, in addition to promoting lipid-specific autophagy, lipophagy [129]. Indeed, lipophagy could contribute to the overall degradation of intramuscular lipid droplets (i.e., by synergizing with the increased activity of cytosolic lipases), whereby their lysosomal export as fatty acids would lead to mitochondrial uptake, oxidation and ATP production to maintain cellular bioenergetics during stress [130].

These observations fit well with how END would elicit an increased autophagic flux to sustain the energetic demands of prolonged contractile activity. Firstly, PGC-1 α itself can transcriptionally regulate TFEB [131], and the coordinated turnover of mitochondria (i.e., undulations of biogenesis and mitophagy) is governed by a TFEB/PGC-1 α autoregulatory loop [132,133]. Secondly, PGC-1 α transgenic expression in skeletal muscle enhances autophagosome formation and increases lysosome number, in turn elevating glycogen storage [134]. It is noteworthy that

the basal levels of PGC-1 α are higher in END-trained versus untrained skeletal muscle and that this positively associates with mitochondrial ATP production [135]. Thus, collectively, the available evidence suggests that the oxidative phenotype conferred by PGC-1 α is partially dependent on autophagy and contributes to the overall mitochondrial ATP yield in highly-trained individuals. From a performance perspective, the higher glycogen availability that accompanies lysosome abundance [134] and a preferential uptake of lipid (via lipophagy) by these lysosomes could contribute to the enhanced END-induced lipolysis of intramuscular lipid in endurance-trained athletes [136–138], thereby facilitating “glycogen sparing” and attenuated lactate production that is associated with endurance performance (i.e., raising the lactate threshold and delaying the onset of fatigue at a given exercise intensity) [139–141].

5. Heterogeneity in exercise adaptation: ‘High’ and ‘low’ responders

Resistance and endurance training induce a myriad of anabolic and metabolic response-adaptations that confer beneficial effects on musculoskeletal and cardiorespiratory health, respectively. However, the magnitude of such training-induced responses is variable and unpredictable between individuals [142,143]. Such variability exists despite participant cohorts presenting similar physical (i.e. height, weight) and personal (i.e. age, gender)

characteristics, fitness levels (i.e. $\dot{V}O_{2peak}$, maximum strength) before undertaking standardised exercise intervention. The initial observations of ‘individual responses’ to exercise training came from the HERITAGE study which reported that improvements in $\dot{V}O_{2peak}$ ranged from zero (i.e., no improvement) to 100% of pre-training values following chronic (4–5 months) endurance training in previously sedentary participants [144]. The term ‘non-responders’ has been coined for participants who show little/no improvements in selected parameters/adaptation responses to a given exercise intervention [143]. However, adaptation responses to exercise are extremely complex involving the interplay of different tissues/organs with a high degree of ‘cross-talk’ to co-ordinate the body’s response to exercise. Thus, it is conceivable that one tissue group or associated physiological variable may show little/no response, while another may have a large (compensatory) response. In this regard, a recent study investigated the adaptive response to 12 and 24 weeks of resistance training in older (> 65 years) men and women and reported that all participants increased one of the main outcome variables (lean body mass, leg extension or press strength, muscle fiber size and/or sit-to-stand test results) but not necessarily all of them, or by the same magnitude [145]. Furthermore, some participants had an increase in one (or more) outcome(s) at the completion of the study (i.e., after 24 weeks) whereas other subjects had robust improvements in the early (first 12 weeks) of the program. Thus, the time-course of adaptation may explain part of the variability in ‘individual responses’ and, with time, such differences may be

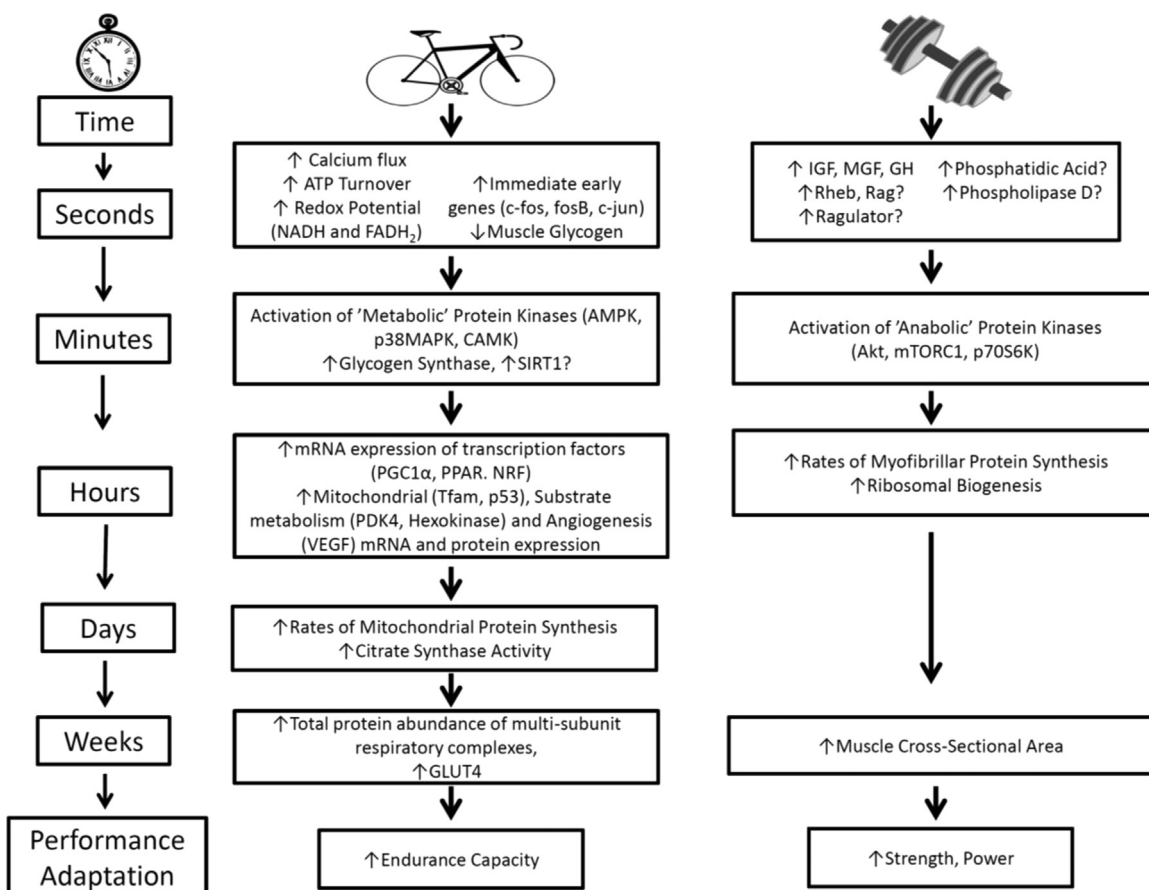


Fig. 3. Time-dependent changes in molecular responses to resistance and endurance exercise in skeletal muscle implicated in enhanced performance. Following endurance exercise, immediate fluctuations in intracellular calcium, ATP and redox potential activate ‘metabolic’ kinases. This is followed by an upregulation in the levels of transcription factors and mitochondrial, substrate and angiogenesis-related mRNAs. These signaling events culminate in mitochondrial biogenesis which results in enhanced substrate metabolism and improved endurance performance. Following resistance exercise, activation of ‘anabolic’ kinases occurs through binding of IGF to its receptor and purported intracellular signaling proteins such as Rheb, Rag and Ragulator. The initiation of initiation and elongation of protein translation form the basis for increases in rates and myofibrillar that, over time, produce increases in muscle mass that result in increased muscular strength and power performance.

unsubstantial. It has been hypothesized the variability in both the magnitude as well as the type (i.e. training outcome) of exercise-induced adaptation is a result of a variation in individual gene sequence regulating the many complex cell signaling networks responsible for driving exercise adaptation responses [142]. An individual's acute gene and protein response to a bout(s) of exercise may therefore underpin, in part, the degree of individual variance when exercise training is performed over the long term (Fig. 3). A brief discussion of the diversity in molecular responses between different individuals that may mediate adaptation processes following resistance and endurance exercise follows. Investigation of such responses, which are often reported but rarely interrogated, may yield important information to help explain the factors that characterise 'high' and 'low' responders to exercise and guide particular individuals to alternate exercise strategies to maximize adaptation responses.

6. Resistance exercise: variability in muscle cell signaling and lean mass gains

Muscle adaptation responses to resistance exercise commonly focus on gains in muscle strength and muscle size although increases in capillary density [146,147], oxidative capacity and mitochondrial protein synthesis [148,149] are also part of the adaptation responses to resistance exercise. Several studies of resistance training have reported remarkable heterogeneity in type I and type II muscle fiber cross-sectional area [150], muscle cross-sectional area [151,152], one-repetition maximum strength [151,152] and isometric strength [152] following chronic (12–16 weeks) training protocols. While numerous factors such as training volume and intensity, rest periods between sets, training status and habitual diet/nutritional interaction are likely to contribute to this variability, there is currently little understanding of the molecular basis for this diversity. Previous studies have implicated changes in anabolic hormones [153], satellite cell activity [154] and myogenic gene expression [150] to underlie the diversity in hypertrophy 'responsiveness' between individuals. It is also plausible that levels of exercise-induced reactive oxygen (ROS) and nitrogen (RNS) species may contribute to heterogeneity in anabolic adaptation responses due to their interaction with cell signaling networks regulating growth factor secretion and muscle inflammation processes [155,156]. Indeed, chronic increases in ROS activity have been shown to attenuate hypertrophy adaptations following isometric contractions suggesting individual gains in muscle mass may be linked to the production, and subsequent buffering, of ROS and RNS within skeletal muscle following resistance exercise [157].

Changes in rates of muscle protein synthesis, and the cell signaling proteins regulating these responses, appear to be the central locus for heterogeneity in hypertrophy adaptations with resistance training. Acute changes in rates of myofibrillar protein synthesis that accompany each training bout are the main locus of control for subsequent increases in muscle fiber size [158]. Regulation of myofibrillar protein synthesis is mediated through a range of diverse, but integrated intracellular processes such as mRNA transcription, protein translation, and several levels of post-translational modifications including DNA methylation, acetylation and histone modification [159]. In regards to resistance exercise-induced increases in myofibrillar protein synthesis, the phosphorylation status and enzyme activity of signaling proteins in the mTORC1 signaling cascade are regularly quantified as proxy markers of muscle protein synthesis following resistance exercise.

Phosphorylation of the protein kinase $p70^{S6K}$ has been associated with muscle hypertrophy responses following resistance training in rodent [160] and human skeletal muscle [24,161].

Mitchell and colleagues observed a significant correlation between $p70^{S6K}$ phosphorylation following an initial resistance exercise bout and gains in mean muscle fiber area after 16 weeks resistance training in young men [24]. This correlation was, in part, found to account for 46% of the variance in the hypertrophic response. Responder status to resistance training, measured solely through increases in muscle fiber area, may therefore be partially dependent on an individual's capacity to acutely induce phosphorylation of this marker. However, this correlation was only apparent with phosphorylation responses observed at 5 h, but not 1 h, post exercise, demonstrating any putative correlations in $p70^{S6K}$ phosphorylation is strictly time-course dependent. Moreover, other chronic resistance training studies have failed to observe this correlation in response with $p70^{S6K}$ [162,163].

The 4E-BP1 has also been shown to correlate with changes in both resistance training-induced increases in muscle volume [164] and rates of myofibrillar protein synthesis [165]. However, similar to the correlative relationship previously identified with $p70^{S6K}$, this association between 4E-BP1 phosphorylation and increases in muscle mass following resistance training were only observable in phosphorylation measured 1 h post an initial resistance exercise session. While these investigations collectively yield new information to the molecular responses involved in anabolic adaptations, there are several shortcomings with using cell signaling responses for determining 'responder' status to resistance training. Firstly, the response of these signaling proteins are time-course dependent, meaning this extremely limited sampling 'snapshot' of protein responses is unlikely to be completely reflective of their true function and activity post-exercise to subsequently mediate anabolic adaptation responses. Secondly, cross-talk between different muscle signaling proteins along with the multiple, diverse cellular functions regulated by certain proteins ultimately means changes observed in the signaling status of a particular protein cannot completely be implicated in its specific role in muscle anabolism with resistance exercise. This may explain why no correlation in mTORC1 phosphorylation with MPS or hypertrophy has been reported when considering its roles in other cell signaling processes such as autophagy [166] and cell differentiation [167] may impact its signaling response to exercise. Finally, phosphorylation status of these signaling markers may not reflect respective activity levels [168] and therefore reduce any putative regulation of the variance in muscle hypertrophy with resistance training.

Compared to the transient changes in cell signaling, quantification of the rate of myofibrillar protein synthesis post-resistance exercise provides greater information to the type and rate at which proteins are being synthesized between designated post-exercise time points to mediate muscle anabolism. Acute changes in MPS following resistance exercise would therefore suggest this response to be extrapolated to muscle hypertrophy and increases in lean body with chronic training. Increases in rates of myofibrillar protein synthesis are routinely observed following acute [169–172] and chronic [173] resistance exercise although not all studies observe an increase [174]. Mayhew and colleagues were first to investigate whether acute changes in rates of MPS following a single bout of resistance exercise correlate with overall increases in lean muscle accrual following chronic (16 weeks) resistance training in young men [175]. Despite elevated rates of MPS 24 h following the initial resistance exercise session, there was no association between changes in MPS and muscle mass indicating acute measures of MPS are not reflective of 'responsiveness' to resistance training. However, a limitation of this study was that participants were fasted when rates of MPS were quantified. A recent study therefore investigated this relationship response when 30 g protein, an amount capable of maximally stimulating MPS, was consumed following an acute bout of

resistance exercise compared to gains in lean mass following 16 weeks resistance training [164]. Previous studies have overwhelmingly shown rates of myofibrillar protein synthesis to be further increased when protein is consumed in the acute recovery period [176–178] and likely forms the mechanistic basis for the findings of a recent meta-analysis concluding protein ingestion to have a positive effect on muscle mass and strength following chronic resistance training compared to a placebo [179]. Despite significant increases in 1–3 h and 3–6 h rates of MPS, there was no correlation in intra-individual changes in acute MPS with changes in muscle volume or lean body mass [164]. While these findings provide little support for acute measures of MPS as surrogate markers for gains in muscle mass and individual hypertrophic potential/responsiveness, several areas of future investigation remain including the length of times in which both acute FSR measures and duration (weeks) of training program, type of resistance training performed (number of sets, repetitions, etc.) and training status of participants at commencement of the training program.

7. Endurance exercise: Heterogeneity in oxidative capacity

An individual's maximal aerobic capacity is a key predictor of metabolic health [180,181]. Indeed, the test for $\dot{V}O_{2\max}$ is now recognized as a better predictor of mortality than any other established risk factor or biomarker for cardiovascular disease [181]. However, there is large heterogeneity ($\sim -2.8\%$ to $\sim 34\%$) in $\dot{V}O_{2\max}$ responses following periods of endurance training [182–184]. This variability in response may be a result of the integration of multiple physiological systems that can limit almost every step of aerobic energy production (cardiac output, oxygen delivery and extraction and muscle oxidative capacity). However, despite the involvement (and limitation) of multiple body tissues, the investigation of molecular markers mediating adaptation responses in skeletal muscle will still yield important information to potential mechanisms underpinning this heterogeneity as such changes in gene sequence will be equally relevant in other participating body tissues [143].

Studies focusing on the molecular responses mediating adaptation responses with endurance training typically focus on cell signaling events increasing the activities of mitochondrial enzymes and total mitochondrial protein abundance that ultimately govern a greater reliance on fat as a fuel source for enhanced prolonged performance. As a result, many studies investigating the cellular events that promote mitochondrial biogenesis focus on PGC-1 α expression due to its coordinated expression of mitochondrial proteins encoded in the nuclear and mitochondrial genomes [2,185].

We recently undertook a study in untrained males ($\dot{V}O_{2\max} \sim 40 \text{ ml kg}^{-1} \text{ min}^{-1}$) to determine if there was any relationship between the individual PGC-1 α mRNA response to an acute, single bout of endurance exercise (60 min at $\sim 70\% \dot{V}O_{2\max}$) and the variation in subsequent endurance capacity following a short-term (8 consecutive days) endurance training program. We found significant training-induced increases in the group for $\dot{V}O_{2\max}$ ($\sim 10\%$, $P=0.002$) and peak power output ($\sim 14\%$, $P=0.01$) following training. Of note, was the large variation in individual responses ranging from no improvement to $\sim 18\%$ (a two-fold increase above the group mean change). PGC-1 α mRNA in response to the acute (pre-training) exercise bout increased 839% ($p < 0.001$) for the group, but ranged from a small decline to a $\sim 1200\%$ increase. We were unable to detect any meaningful association between the changes in PGC-1 α mRNA abundance in response to an acute unfamiliar bout of exercise and the subsequent endurance training-induced improvements in whole-

body $\dot{V}O_{2\max}$.

Using more sophisticated techniques, many studies have used genomic screening in an attempt to investigate heterogeneity in aerobic-based adaptation responses with endurance exercise. Timmons and colleagues used RNA expression profiling to show over 800 gene transcripts, (the 'training-responsive transcriptome') are up-regulated in response to 6 weeks of endurance training [186] and subsequently determined that at least 100 of these 800 genes were differentially regulated between subjects who demonstrated a 'high' and 'low' response with regard to improvements in aerobic capacity [187]. A discussion on the different genomic screening technologies and associated bioinformatics tools to link changes in the muscle genome with physiological responses is beyond the scope of this review.

8. Skeletal muscle signaling and exercise performance: A disconnect

From the preceding discussion, it is clear that exercise performance is a complex phenomenon resulting from the integration of multiple physiological, biomechanical and psychological factors. As such, it is naive to think that any single 'molecular marker' can predict or explain variability in exercise responses and subsequent performance capacity. Indeed, there is often a mismatch between the changes in cellular 'mechanistic' variables (often reported as increases in the phosphorylation status of signaling molecules and/or increases in the expression of genes and proteins involved in mitochondrial biogenesis or muscle protein synthesis) and whole body functional outcomes (changes in training capacity or measures of performance). There are likely to be several reasons for the apparent disconnect. First, and foremost, there may be no direct relationship between performance and some of the training-induced changes in selected cellular events that have been measured! Just because we can measure a new signaling protein or some of its downstream kinases, does not mean it has an important role in exercise performance. While two signaling proteins, the AMPK and PGC-1 play major roles in endurance-training adaptation, it should be noted that normal responses and adaptations to both acute exercise and chronic exercise training can be seen when one or more key pathways are absent, are blocked with drugs, or are otherwise attenuated. The functions achieved by up-regulating various muscle proteins may be permissive in promoting the capacity for exercise but are not quantitatively correlated, or indeed rate-limiting for athletic performance. Indeed, skeletal muscle function is only one factor in determining performance. Of course, it may be simply that we currently lack the appropriate tools to accurately measure exercise/sports performance, in particularly the ability to detect small changes that are worthwhile to a competitive athlete in order to change the outcomes of real world events. In some instances, the technical variability of various enzymatic assays and/or gene measurements is likely to exceed the small biological changes that manifest as improvements in performance. Many of the studies that involve training interventions and measure a variety of 'molecular markers' and subsequent performance capacity use moderately-trained individuals and whether highly-trained athletes have a different response or require a different stimulus to untrained or even moderately trained individuals warrants further investigation. In this regard, Ljubicic and Hood [188], reported an attenuation of kinase phosphorylation in muscle with high mitochondrial content and suggests that these proteins may require a greater stimulus input for activation to propagate these cues downstream to evoke phenotypic adaptations. This phenomenon was first reported by Yu et al. [189].

9. Conclusion

Over the past 50 years, there have been major advances in our understanding of the molecular pathways involved in the exercise response-adaptation. Various “omics” technologies and the application of computational and systems biology to exercise bioenergetics has facilitated progress in unraveling the complexity of signaling networks involved in the exercise response. It has become clear that the breadth of signaling pathways and kinases modulated by contraction is far greater than previously appreciated. For example, a recent global phosphoproteomic analysis of human muscle in response to a single bout of intense exercise (10 min cycling at $\sim 90\%$ of $\dot{V}O_{2\max}$) revealed ~ 1000 unique exercise-regulated phosphosites on ~ 550 proteins, with many of these targets never before implicated in exercise signaling [190]. Such approaches to unraveling the complexity of exercise-induced signaling events is both exciting and somewhat daunting because it is already apparent that there are many redundant molecular pathways engaged in many key acute and chronic responses to exercise. Indeed, direct evidence linking specific signaling events to defined performance measures or specific health outcomes and understanding how these effects exert their benefits in different populations is elusive and represents a major challenge for future research in this field. Connecting distinct signaling cascades to defined metabolic responses and specific changes in gene expression in skeletal muscle that occur after exercise is likely to be a complex and perhaps, ultimately futile task because the majority of these pathways are not linear, but, rather, they constitute a complex network, with a high degree of crosstalk, feedback regulation, and transient activation. In the final analyses, perhaps the only obligatory response to the numerous and widespread exercise-induced perturbations to cellular homeostasis is the defense of homeostasis itself.

Conflict of interest

The authors indicate no potential conflicts of interest.

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